Jingze Dai

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SKILLS AND CAPABILITIES (Front-End | Back-End | Full-Stack | DevOps | CI/CD | Machine Learning)

Technical Skills: **Advanced in** MongoDB, XML, OpenGL, OpenCV, AWS/GCP, DB2, Jenkins, Azure, GraphQL, Agile, R, SDLC, PyTorch/TensorFlow, Typescript, PostgreSQL, Docker, Golang; **Proficient in** Python, Java, C/C++, C#, Latex, .NET, Haskell, jQuery, Next.js, Spring, JavaScript, HTML, CSS, Ruby, Oracle, PostgreSQL/MySQL, Scala, JSON, Ansible, PowerBI, Kubernetes, SciPy/NumPy, React, Unity, Kotlin, git, Ubuntu, Linux/Unix, Android/MacOS, Swift, Rust, Xcode, MS Word, PowerShell, Excel, Scikit-Learn, Vue.js, TCP/IP, Selenium, Groovy, OpenAI (GPT), NoSQL, Apache (Hadoop & Spark), ETL.
Certifications: IBM Data Science, Machine Learning Specialization (Coursera), SQL **Advanced** & Angular, Rest APIs, Node.js **Intermediate** Certificate (HackerRank), Salesforce Development Professional, SAS **Advanced**, Deep Learning A-Z (Udemy).

WORK EXPERIENCE

Software Research Assistant — Jan 2024 - Present
Cornell University & McMaster University — Ithaca, NY
- Wrote 6 conditions' stochastic energy systems' simulation models, and conducted independent proofs for each part, which determined the formulas of optimal parameters triggering the minimal two-layer cost. (Submitted to **LOCO 2024, Paper 27**)
- Devised 20 image-transformed CNN methods to detect V2X misbehavior messages, reaching a 20-class detection accuracy of 98% and a detection time within 0.06 second, using explainable AI methods to select 3 most important driving features.
- Read 132 recent IoT publications, and summarized a survey indicating recent XAI applications. (Displayed at **ISICN 2025**)

Computer Science Course Teaching Assistant (CS 1XC3 & CS 3SH3) — May 2023 - Dec 2023
McMaster University — Hamilton, ON
- Crafted 16 categories of operating system functions, taught C programming and LaTeX documentation to 200+ students, which achieved a 95% positive feedback rate for clarity and effectiveness.
- Evaluated students' 32 C code assignments and lab projects about multithreading and caching, ensuring fair grading and providing detailed feedback that clarified 110+ points of confusion for students.

Software Developer (Coop) — Sept 2021 - Apr 2022
Government of Ontario, CYSSC services (Ministry of Children, Community, and Social Services) — Toronto, ON
- Developed 5 different MCCSS service websites, and resolved 3 existing bugs and errors. Incorporated login, questionnaire and appointment functions on the webpage, with 870 users more each month.
- Applied Angular, MySQL, Python, and Java to build an automated tester to test the website (login pages and fingerprint checks). Developed over 220 test cases, which showed 18 errors and exceptions. Mitigated all mentioned errors.

EDUCATION

Statistics, Cornell University — Ithaca, NY
Master of Science (M.Sc.) — Sept 2025 - Present
Major in Statistics, with specialization in **Data Science**
Relevant Coursework: Algorithms, Data Structures, Data Mining, Machine Learning, Artificial Intelligence, Natural Language Processing, Image Processing, Cloud/High-Performance Computing, Database Systems, Stochastic Calculus, Large Language Models, Time Series/Multivariate Analysis, Big Data Management, Robotics, Quantum Computing, Large-Language Models.

PROJECTS

MuseScore - Music notation and composition (https://github.com/musescore/MuseScore) — Dec 2022 - Present
- Implemented 3 new mouse cursor shape effects when dragging 3 different movable panels and sections.
- Engineered 2 novel approaches to insert braille panels on the view menu, and designed 1 hotkey to enable/disable the braille panel. Inserted 2 optional arrows to increase the braille function visibility.

Drasil - Generate All the Things (https://github.com/JacquesCarette/Drasil) — Jan 2022 - Present
- Mitigated 25 issues from Haskell automatic generations, improved 32 existing code segments with expected format.
- Added 8 group's references and constraint components the SRS documentation, added 20+ components to its website.
- Inspected 7 tricky and complicated execution errors' messages, and discovered their reasons and fixing methods.

Algorithm Design Implementations (https://github.com/daijingz/Algorithm-and-Design) — May 2021 - Present
- Wrote over 70 kinds of algorithm implementations programs in Python, Java, C++, C, Haskell. (With 36 unit testing parts)
- Constructed 8 application projects applying some algorithms, improving at least 30% performance